

Voice-Based Concept Understanding Analyser (VBCUA)

Project Description:

The **Voice-Based Concept Understanding Analyser (VBCUA)** is an AI-powered educational application designed to evaluate a student's conceptual understanding through spoken explanations. Instead of relying solely on written assessments, the system enables students to verbally explain a concept, allowing educators to assess both conceptual knowledge and communication effectiveness.

The platform integrates **OpenAI Whisper** for speech-to-text transcription, **Sentence-BERT** for semantic similarity analysis, and **Librosa** for extracting speech characteristics such as pause ratio, RMS energy, speech duration, and waveform visualization. These metrics are combined within a scoring engine to calculate an overall understanding score and classify the student's performance as **Strong Understanding, Moderate Understanding, or Poor Understanding**.

Developed using **Python** and **Streamlit**, the application provides an intuitive interface for uploading audio files, viewing transcription results, visualizing audio waveforms, and downloading comprehensive PDF evaluation reports. With efficient AI model caching, modular architecture, and secure processing, VBCUA offers educators and learners an intelligent solution for assessing conceptual understanding through spoken responses.

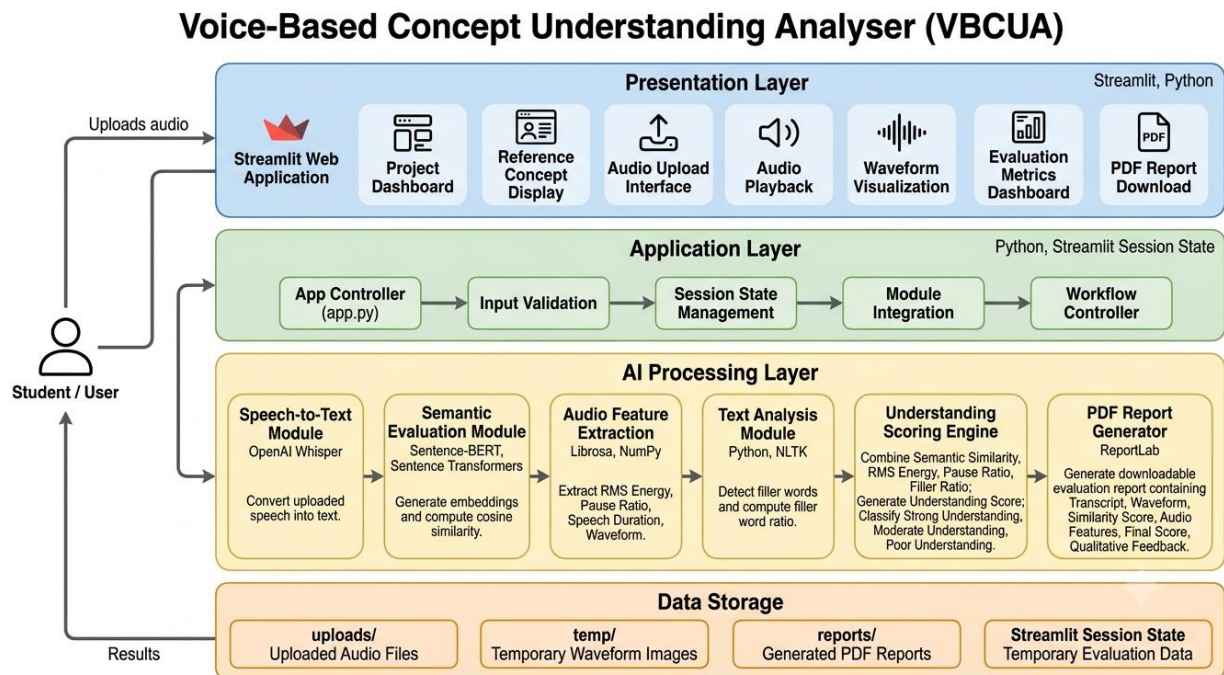
Scenarios

Scenario 1: A student explains the concept of **Machine Learning** by uploading a recorded audio explanation in WAV format. The system converts the speech into text using OpenAI Whisper, compares the explanation with the reference concept using Sentence-BERT, extracts audio features, computes an understanding score, and generates a downloadable PDF report summarizing the evaluation.

Scenario 2: An instructor assigns students to explain **Artificial Intelligence** concepts verbally. Each student's audio explanation is evaluated automatically for semantic similarity, filler word usage, pause ratio, and confidence metrics. The application classifies each student's conceptual understanding and produces personalized feedback.

Scenario 3: A learner preparing for an oral examination uploads multiple practice recordings over time. The application evaluates each explanation independently, enabling the learner to identify improvements in conceptual accuracy, fluency, and communication confidence before the final assessment.

Technical Architecture:



Description: The Voice-Based Concept Understanding Analyser follows a modular three-layer architecture to deliver efficient AI-powered spoken concept evaluation. The Presentation Layer is developed using Streamlit, providing an interactive web interface where users upload audio recordings, view waveform visualizations, analyze evaluation metrics, and download PDF reports. The Application Layer manages application workflow, validates uploaded audio, coordinates communication between processing modules, maintains session state, and integrates all evaluation components. The AI Processing Layer combines OpenAI Whisper for speech transcription, Sentence-BERT for semantic similarity computation, Librosa for audio feature extraction, and a custom scoring engine for concept evaluation. ReportLab is used to generate professional PDF reports containing evaluation summaries and qualitative feedback. The application stores uploaded audio files, generated waveform images, and PDF reports within local project directories for efficient processing and retrieval.

Pre-requisites:

- Python Programming
- Streamlit Framework
- OpenAI Whisper
- Sentence Transformers (Sentence-BERT)
- Librosa & SoundFile
- NumPy & Matplotlib
- ReportLab
- NLTK
- Visual Studio Code
- Virtual Environment (venv)
- Git & GitHub (Optional for Version Control)

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Set Up the Development Environment.
- **Activity 1.2:** Research and Select Appropriate AI Models
- **Activity 1.3:** Define the System Architecture
- **Activity 1.4:** Configure the Development Environment

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the Speech-to-Text Module
- **Activity 2.2:** Develop the Semantic Evaluation Module
- **Activity 2.3:** Develop the Audio Feature Extraction Module
- **Activity 2.4:** Develop the Text Analysis Module
- **Activity 2.5:** Develop the Understanding Scoring Engine
- **Activity 2.6:** Develop the PDF Report Generator

Milestone 3: Application Development

- **Activity 3.1:** Develop app.py
- **Activity 3.1:** Session Management

Milestone 4: Frontend Development

- **Activity 4.1:** User Interface Design
- **Activity 4.2:** Dynamic Result Display
- **Activity 4.3:** Error Handling

Milestone 5: Testing and Deployment

- **Activity 5.1:** Functional Testing
- **Activity 5.2:** Performance Optimization
- **Activity 5.3:** Deployment Preparation

Milestone 6: Conclusion

Milestone 1: Model Selection and Architecture

Activity 1.1: Set Up the Development Environment

Before implementing the application, configure the Python environment and install all required dependencies.

Steps

- Create a dedicated Python virtual environment.
- Install all project dependencies from requirements.txt.
- Verify successful installation of Whisper, Sentence-BERT, Librosa, Streamlit, ReportLab, and supporting libraries.
- Confirm compatibility with Python 3.10+.
- Test library imports before development.

Activity 1.2: Research and Select Appropriate AI Models

Research suitable AI models capable of performing speech transcription and semantic similarity analysis.

The following considerations should be evaluated:

- Understand the requirements for spoken concept evaluation.
- Compare speech recognition models for transcription accuracy.
- Evaluate semantic embedding models for conceptual similarity measurement.
- Analyze computational efficiency and inference speed.
- Select OpenAI Whisper for speech transcription.
- Select Sentence-BERT (all-MiniLM-L6-v2) for semantic similarity analysis.

- Use Librosa for speech feature extraction including pause ratio, RMS energy, and waveform generation.

Activity 1.3: Define the System Architecture

Design a modular architecture illustrating communication between the frontend, backend, AI models, and reporting components.

The architecture should include:

- Streamlit Presentation Layer
- Audio Upload Module
- Speech-to-Text Module (Whisper)
- Semantic Evaluation Module (Sentence-BERT)
- Audio Feature Extraction Module (Librosa)
- Understanding Scoring Engine
- PDF Report Generator
- Local Storage for Audio Files and Reports

Activity 1.4: Configure the Development Environment

Prepare the application development environment by:

- Installing Python packages using requirements.txt.
- Configuring project folders (uploads, reports, temp, tests).
- Setting up Visual Studio Code.
- Testing Whisper model loading.

- Testing Sentence-BERT embedding generation.
- Validating audio processing using Librosa.
- Verifying Streamlit application startup.

Milestone 2: Core Functionalities Development

In this milestone, the core AI modules of the Voice-Based Concept Understanding Analyser are developed. The application integrates speech recognition, semantic analysis, audio feature extraction, and intelligent scoring to evaluate students' conceptual understanding from spoken explanations.

Activity 2.1: Develop the Speech-to-Text Module

The Speech-to-Text module converts uploaded audio recordings into textual transcripts using OpenAI Whisper.

Tasks Performed

- Load the Whisper model using Streamlit resource caching.
- Accept uploaded WAV audio files.
- Normalize audio input before transcription.
- Convert speech into text.
- Handle transcription errors gracefully.
- Validate transcription quality across different speech styles and recording conditions.

Technology Used

- OpenAI Whisper
- Python
- Streamlit

Activity 2.2: Develop the Semantic Evaluation Module

The Semantic Evaluation module compares the student's explanation with the reference concept to determine conceptual similarity.

Tasks Performed

- Generate sentence embeddings using Sentence-BERT.
- Generate embeddings for the reference concept.
- Compute cosine similarity.
- Normalize similarity scores.
- Classify conceptual similarity levels.

Technology Used

- Sentence-BERT
- Sentence Transformers
- Scikit-learn

Activity 2.3: Develop the Audio Feature Extraction Module

This module analyzes speech quality using Librosa.

Audio Features Extracted

- Speech Duration
- RMS Energy
- Pause Ratio

- Waveform Visualization

Tasks Performed

- Load uploaded audio.
- Extract acoustic features.
- Generate waveform visualization.
- Save waveform image for display and PDF generation.

Technology Used

- Librosa
- NumPy
- Matplotlib
- SoundFile

Activity 2.4: Develop the Text Analysis Module

The transcript generated by Whisper is analyzed for communication quality.

Tasks Performed

- Clean transcribed text.
- Detect filler words.
- Calculate filler word ratio.
- Compute transcript statistics.

Technology Used

- Python
- NLTK
- Regular Expressions

Activity 2.5: Develop the Understanding Scoring Engine

The scoring engine combines multiple evaluation metrics into a final understanding score.

Evaluation Metrics

- Semantic Similarity
- Filler Word Ratio
- Pause Ratio
- RMS Energy

Output

- Final Understanding Score
- Strong Understanding
- Moderate Understanding
- Poor Understanding

Technology Used

- Python

Activity 2.6: Develop the PDF Report Generator

Generate a professional evaluation report summarizing the analysis.

Report Includes

- Reference Concept
- Student Transcript
- Audio Waveform
- Semantic Similarity
- Audio Features
- Final Score
- Understanding Level
- Qualitative Feedback

Technology Used

- ReportLab

Milestone 3: Application Development

During this milestone, all backend modules are integrated into a unified Streamlit application.

Activity 3.1: Develop app.py

The main application controls the complete evaluation workflow.

Tasks Performed

- Create Streamlit interface.

- Upload audio files.
- Display reference concept.
- Call Speech-to-Text module.
- Perform semantic evaluation.
- Extract audio features.
- Compute understanding score.
- Display evaluation metrics.
- Generate waveform visualization.
- Generate downloadable PDF report.

Activity 3.2: Session Management

Tasks Performed

Use Streamlit Session State to store:

- Uploaded Audio
- Transcript
- Semantic Similarity
- Audio Features
- Understanding Score
- Evaluation Results
- Generated PDF

This avoids recomputing results during interaction.

Milestone 4: Frontend Development

This milestone focuses on designing a responsive and user-friendly interface.

Activity 4.1: User Interface Design

Develop a clean Streamlit interface consisting of:

- Project Header
- Reference Concept
- Audio Upload Section
- Audio Playback
- Waveform Visualization
- Analysis Button
- Evaluation Summary
- Download PDF Button

Activity 4.2: Dynamic Result Display

Display analysis results dynamically after processing.

Display Components

- Student Transcript
- Semantic Similarity
- Pause Ratio
- RMS Energy
- Filler Word Ratio
- Understanding Score
- Classification

- Qualitative Feedback

Activity 4.3: Error Handling

Implement validation for:

- Missing audio files.
- Unsupported audio formats.
- Corrupted uploads.
- Empty transcriptions.
- AI processing failures.

Milestone 5: Testing and Deployment

The application is tested and optimized before deployment.

Activity 5.1: Functional Testing

Test all application modules.

Modules Tested

- Audio Upload
- Speech Transcription
- Semantic Evaluation
- Audio Feature Extraction
- Filler Word Detection
- Scoring Engine
- Waveform Generation

- PDF Report Generation

Activity 5.2: Performance Optimization

Optimize execution using:

- Streamlit Resource Caching
- Efficient Model Loading
- Optimized Audio Processing
- Reduced Memory Usage

Activity 5.3: Deployment Preparation

Prepare the application for deployment.

Tasks Performed

- Verify requirements.txt
- Validate dependencies
- Test on local machine
- Configure Streamlit Community Cloud
- Verify application startup
- Perform end-to-end testing

Milestone 6: Conclusion

The **Voice-Based Concept Understanding Analyser (VBCUA)** successfully integrates Artificial Intelligence, Natural Language Processing, and Audio Signal Processing to assess students' conceptual understanding through spoken explanations.

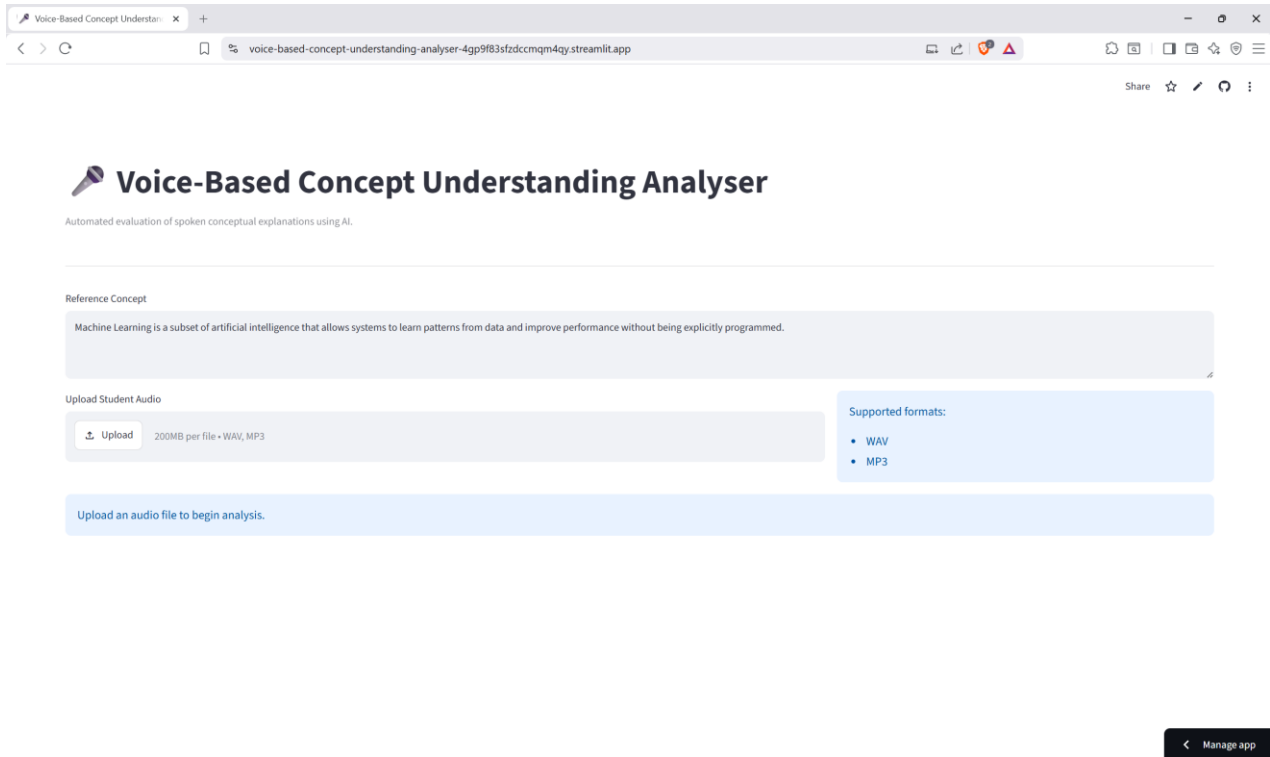
The system combines **OpenAI Whisper**, **Sentence-BERT**, **Librosa**, and a custom evaluation engine to provide an automated and objective assessment of conceptual knowledge and communication quality.

The application demonstrates the practical use of AI in educational assessment by generating accurate transcriptions, semantic similarity scores, speech quality metrics, understanding classifications, and professional PDF reports through a responsive Streamlit interface.

Future enhancements may include multilingual support, real-time speech evaluation, cloud database integration, user authentication, teacher dashboards, and historical performance analytics.

Exploring the Website's Web Pages:

Home / Generator Page:



The screenshot shows a web browser window with the address bar displaying "voice-based-concept-understanding-analyser-4gp9f83sfzdcqm4qy.streamlit.app". The page title is "Voice-Based Concept Understanding Analyser" with a microphone icon. Below the title is the subtitle "Automated evaluation of spoken conceptual explanations using AI." The main content area is divided into two sections: "Reference Concept" and "Upload Student Audio". The "Reference Concept" section contains a text box with the text: "Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed." The "Upload Student Audio" section features an "Upload" button with a file icon, a text label "200MB per file • WAV, MP3", and a list of "Supported formats" including "WAV" and "MP3". A light blue banner at the bottom of the upload section says "Upload an audio file to begin analysis." In the bottom right corner, there is a "Manage app" button.

Description: The single-page interface of the Voice-Based Concept Understanding Analyser (VBCUA) serves as both the landing page and the analysis workspace. The application features a modern, responsive interface built with Streamlit, providing users with an intuitive environment for evaluating spoken conceptual explanations. The page displays the application title, a brief description of its purpose, and a predefined reference concept that serves as the basis for semantic evaluation. The main interface integrates audio upload controls, waveform visualization, analysis controls, evaluation metrics, and PDF report generation into a seamless workflow.

Audio Upload Section:

Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

Reference Concept

Sample Audio

Upload Student Audio

 Sample_...46e458.mp3
0.5MB

Supported formats:

- WAV
- MP3

0:01 / 0:27

Analyze Concept Understanding

Analysis Completed

Description:

The Audio Upload section enables users to upload their spoken explanations in supported audio formats (WAV). Users can review the selected audio using the built-in playback controls before initiating analysis. Once an audio file is uploaded, the application validates the input and prepares it for processing. Clicking the Analyze Concept Understanding button starts the AI evaluation pipeline, during which a loading spinner indicates that transcription, semantic analysis, audio feature extraction, and scoring are in progress.

Analysis Results Section:

Transcribed Explanation

Artificial intelligence is a technology that allows computers to perform tasks that normally requires human intelligence. Those tasks include learning, reasoning, problem solving, understanding languages and recognize image, A used to in virtual assistants, chatbots, autonomous vehicles, healthcare and many other industries. Artificial intelligence helps improve efficiency and automate complex tasks.

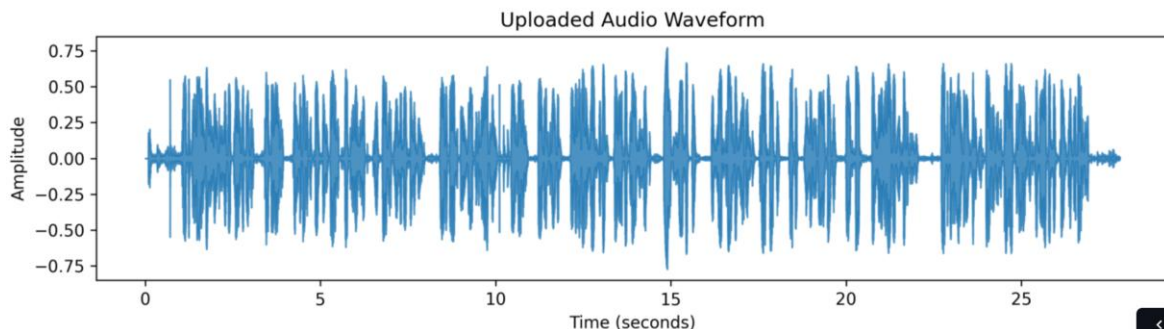
Final Evaluation

Understanding Score

55/100

Poor Understanding

Audio Visualization



< Manage app

Description:

After successful processing, the application dynamically displays the evaluation results. The results page presents the automatically generated speech transcript alongside the predefined reference concept for comparison. Audio waveform visualization is rendered to provide transparency into the uploaded recording.

The evaluation summary displays key performance metrics, including Semantic Similarity Score, Filler Word Ratio, Pause Ratio, RMS Energy (Confidence Metric), Final Understanding Score, and the qualitative understanding classification (Strong, Moderate, or Poor Understanding). Users can also download a professionally formatted PDF report containing the complete evaluation, waveform image, metric tables, and personalized feedback.

Conclusion

The Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered educational application developed using Python, Streamlit, OpenAI Whisper, Sentence-BERT, Librosa, and ReportLab to evaluate conceptual understanding from spoken explanations. The application combines speech recognition, semantic similarity analysis, and audio feature extraction to generate objective understanding scores and qualitative feedback.

The project successfully integrates multiple artificial intelligence and signal processing techniques into a unified educational assessment platform. Its modular architecture ensures efficient processing while providing an intuitive user experience through interactive visualization and automated PDF report generation.

Looking ahead, the system offers several opportunities for enhancement, including support for multiple languages, real-time speech evaluation through microphone input, cloud-based storage for maintaining student assessment history, teacher analytics dashboards, user authentication, and advanced AI-driven feedback generation. With these future improvements, the Voice-Based Concept Understanding Analyser has the potential to evolve into a comprehensive intelligent learning assessment platform for educational institutions and online learning environments.